
Semi One-Class Landmark-Only Liveness Detection via Frequency-Split Variational Modeling

Dong min Kim ^{*1} Il young Kim ^{*2} Sang june Ahn ^{*3}

Abstract

Face liveness detection is critical for preventing presentation attacks in face recognition systems. However, existing methods often require extensive labeled fake samples and computationally expensive deep networks. We propose an efficient two-stage approach that combines frequency-decoupled variational autoencoders (VAE) with landmark-only input. In Stage 1, we perform one-class learning on real samples using a novel band-split VAE that decomposes facial motion into interpretable frequency bands. Stage 2 fine-tunes the model with a small number of fake samples using either margin-based or discriminative losses. By operating solely on facial landmarks (40 points), our method achieves 89.26% F1-score with significantly fewer parameters than video-based approaches. The frequency decomposition provides interpretability, revealing that fake videos exhibit distinct anomalies in different frequency bands. Our approach demonstrates that effective liveness detection is achievable with minimal supervision and computational resources.

1. Introduction

Face recognition systems are increasingly vulnerable to presentation attacks, where adversaries use photos, videos, or deepfake-generated content to spoof authentication systems. (Yu et al., 2021; Kaur et al., 2024) Liveness detection—distinguishing live faces from fake presentations—has become essential for securing biometric systems.

Traditional liveness detection methods rely on hand-crafted features such as texture analysis or motion patterns. Recent deep learning approaches achieve higher accuracy

but require: (1) extensive labeled datasets of both real and fake samples, (2) computationally expensive 3D CNNs or recurrent networks operating on full video frames, and (3) substantial model parameters. (Bohacek & Farid, 2024)

We identify three fundamental limitations in existing approaches. First, **data efficiency**: most methods require large-scale labeled fake datasets, which are expensive to collect and quickly become outdated as attack methods evolve. Second, **computational efficiency**: processing full video frames with deep networks incurs high computational costs. Third, **interpretability**: deep models provide limited insight into *why* a sample is classified as fake an issue that has also been emphasized in recent deepfake video detection surveys (Yu et al., 2021; Kaur et al., 2024).

In this work, we propose a novel two-stage framework that addresses these limitations through three key contributions:

(1) Landmark-only representation: We operate exclusively on 40 facial landmark trajectories rather than raw pixels, reducing input dimensionality from $300 \times H \times W \times 3$ to $300 \times 40 \times 2$ (where 300 is temporal length). This achieves 1000 times parameter reduction while preserving essential motion dynamics.

(2) Efficient two-stage learning: Stage 1 employs one-class learning on real samples only, learning the manifold of genuine facial motion through a variational autoencoder. Stage 2 fine-tunes with minimal fake samples (538 vs. 1088 real) using either margin-based or discriminative objectives, demonstrating that extensive fake datasets are unnecessary.

(3) Frequency-decoupled architecture: We introduce a band-split VAE that decomposes facial motion into three interpretable frequency bands—low (0–2Hz), band-pass (2–8Hz), and high (8–15Hz)—corresponding to head pose, speech-related motion, and micro-expressions. This decomposition enables interpretability and reveals that fake videos exhibit characteristic anomalies in specific frequency bands.

Our experiments on 234 test samples demonstrate that both proposed Stage 2 methods achieve strong performance: Method 1 (Margin) obtains 88.89% F1-score and Method 2 (Discriminator) achieves 89.26% F1-score, both signif-

^{*}Equal contribution ¹Department of Health and Environmental Science, Korea University ²Department of Computer Science, Korea University ³Department of Linguistics, Korea University. Correspondence to: <>.

icantly outperforming the Stage 1 baseline (86.64% F1) with only 538 additional fake training samples. The frequency decomposition provides actionable insights: fake videos show anomalies primarily in band-pass frequencies, while replayed videos exhibit uniform degradation across all bands.

2. Method

2.1. Overview

Our framework consists of two stages: (1) one-class VAE pretraining on real samples, and (2) fine-tuning with minimal fake samples. The key innovation is a frequency-decoupled architecture that processes three separate frequency bands independently.

2.2. Input Representation

Given a video sequence of length $T = 300$ frames at 30 fps (10 seconds), we extract 40 facial landmarks per frame using a pre-trained detector. Let $\mathbf{L}_t \in \mathbb{R}^{40 \times 2}$ denote landmarks at frame t , where each point is normalized by image dimensions. We construct a feature tensor $\mathbf{X} \in \mathbb{R}^{T \times 40 \times F}$ where $F = 8$ includes:

- Position: (x, y) coordinates
- Velocity: $\mathbf{v}_t = (\mathbf{L}_t - \mathbf{L}_{t-1}) \cdot \text{fps}$
- Acceleration: $\mathbf{a}_t = (\mathbf{v}_t - \mathbf{v}_{t-1}) \cdot \text{fps}$
- Angle: $\theta_t = \arctan 2(y, x)$
- Angular velocity: $\dot{\theta}_t = (\theta_t - \theta_{t-1}) \cdot \text{fps}$

2.3. Band-Split VAE Architecture

We apply 4th-order Butterworth filters to decompose $\mathbf{X}$ into three frequency bands:

$$\mathbf{X}_{\text{LF}} = \text{LowPass}(\mathbf{X}, f_c = 2\text{Hz}) \quad (1)$$

$$\mathbf{X}_{\text{BP}} = \text{BandPass}(\mathbf{X}, [2, 8]\text{Hz}) \quad (2)$$

$$\mathbf{X}_{\text{HF}} = \text{HighPass}(\mathbf{X}, f_c = 8\text{Hz}) \quad (3)$$

Figure 1 illustrates the overall architecture.

Each band is processed by an independent VAE with shared architecture:

Encoder: For band $b \in \{\text{LF}, \text{BP}, \text{HF}\}$, the encoder maps $\mathbf{X}_b \in \mathbb{R}^{C_{\text{in}} \times T}$ (where $C_{\text{in}} = 40 \times 8 = 320$) to latent parameters:

$$\mathbf{h}_b = \text{TCN}_{\text{enc}}(\mathbf{X}_b; C_h = 48, \text{dilations} = [1, 2, 4]) \quad (4)$$

$$[\boldsymbol{\mu}_b, \log \sigma_b^2] = \text{Conv1D}(\mathbf{h}_b) \in \mathbb{R}^{2C_z \times T} \quad (5)$$

where TCN denotes a temporal convolutional network with dilated convolutions, $C_h = 48$ hidden channels, and $C_z = 12$ latent dimensions.

Decoder: Latent codes are sampled via reparameterization $\mathbf{z}_b = \boldsymbol{\mu}_b + \sigma_b \odot \epsilon$ where $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, then decoded:

$$\hat{\mathbf{X}}_b = \text{TCN}_{\text{dec}}(\mathbf{z}_b; C_h = 48, \text{dilations} = [4, 2, 1]) \quad (6)$$

The complete model has three independent VAEs with shared hyperparameters but separate weights, totaling $\sim 2.7\text{M}$ parameters (vs. $>100\text{M}$ for video-based 3D CNNs).

2.4. Stage 1: One-Class Pretraining

Stage 1 trains exclusively on real samples $\{\mathbf{X}_{\text{real}}^{(i)}\}_{i=1}^{N_{\text{real}}}$ with the standard VAE loss:

$$\mathcal{L}_{\text{VAE}} = \sum_{b \in \{\text{LF}, \text{BP}, \text{HF}\}} \left[\|\mathbf{X}_b - \hat{\mathbf{X}}_b\|_1 + \beta \cdot D_{\text{KL}}(\mathcal{N}(\boldsymbol{\mu}_b, \sigma_b^2) \parallel \mathcal{N}(0, \mathbf{I})) \right] \quad (7)$$

where $\beta = 1.0$ for all bands, and D_{KL} denotes the KL divergence. We use L1 reconstruction loss because it is more robust to outliers than L2.

This stage learns the manifold of genuine facial motion. Real samples exhibit:

- **LF band:** Smooth head pose changes
- **BP band:** Periodic speech-related motion
- **HF band:** Natural micro-expressions

2.5. Stage 2: Discriminative Fine-tuning

Stage 2 introduces fake samples $\{\mathbf{X}_{\text{fake}}^{(j)}\}_{j=1}^{N_{\text{fake}}}$ where $N_{\text{fake}} = 538 \ll N_{\text{real}} = 1088$. We explore two approaches:

2.5.1. METHOD 1: MARGIN-BASED LOSS

We encourage fake samples to have reconstruction loss above a margin m :

$$\mathcal{L}_{\text{margin}} = \mathbb{E}_{\mathbf{X}_{\text{real}}}[\mathcal{L}_{\text{VAE}}(\mathbf{X})] + \mathbb{E}_{\mathbf{X}_{\text{fake}}}[\max(0, m - \mathcal{L}_{\text{VAE}}(\mathbf{X}))] \quad (8)$$

where $m = 0.5$ is the target margin. This objective pushes fake samples away from the real manifold while maintaining tight reconstruction for real samples.

2.5.2. METHOD 2: LATENT DISCRIMINATOR

We add a discriminator D_ϕ operating on concatenated latent codes:

$$\mathbf{z}_{\text{concat}} = [\text{pool}(\boldsymbol{\mu}_{\text{LF}}); \text{pool}(\boldsymbol{\mu}_{\text{BP}}); \text{pool}(\boldsymbol{\mu}_{\text{HF}})] \quad (9)$$

$$\text{pool}(\boldsymbol{\mu}) = [\text{mean}(\boldsymbol{\mu}); \text{max}(\boldsymbol{\mu})] \in \mathbb{R}^{2C_z} \quad (10)$$

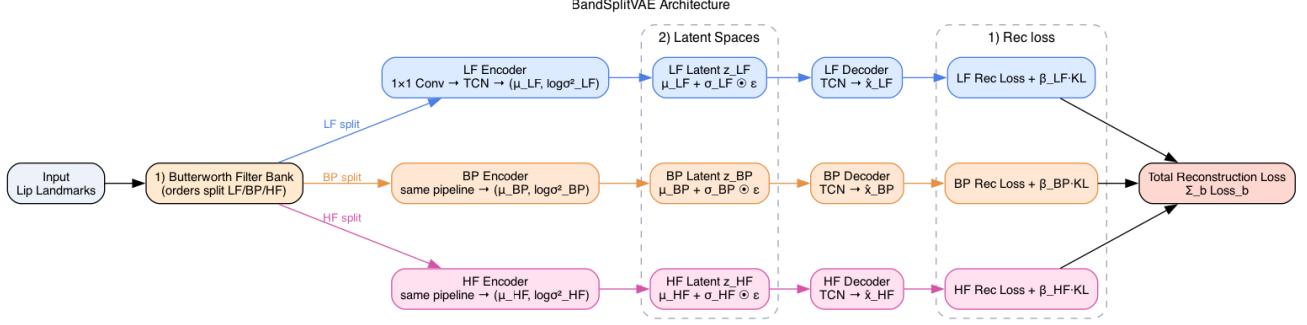


Figure 1. Band-Split VAE Architecture. Input landmarks are decomposed into three frequency bands (LF, BP, HF), each processed by an independent VAE with shared hyperparameters but separate weights.

where pooling aggregates temporal information. The discriminator is a 3-layer MLP with 128 hidden units:

$$p_{\text{fake}} = \sigma(D_{\phi}(\mathbf{z}_{\text{concat}})) \quad (11)$$

The total loss combines VAE reconstruction and binary classification:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{VAE}} + \lambda \cdot \mathcal{L}_{\text{BCE}}(D_{\phi}(\mathbf{z}), y) \quad (12)$$

where $y \in \{0, 1\}$ is the real/fake label and $\lambda = 1.0$. This approach jointly optimizes reconstruction and discrimination.

2.6. Inference: Two-Stage Gaussian Filtering

At inference, we use a statistical approach based on Gaussian modeling of real samples from Stage 2 training data:

Stage 1 Filter (Reconstruction): For each band, we compute reconstruction loss:

$$\ell_b = \|\mathbf{X}_b - \hat{\mathbf{X}}_b\|_2^2, \quad b \in \{\text{LF, BP, HF}\} \quad (13)$$

We form a 3D loss vector $\ell = [\ell_{\text{LF}}, \ell_{\text{BP}}, \ell_{\text{HF}}]^T$ and fit a multivariate Gaussian $\mathcal{N}(\mu_{\ell}, \Sigma_{\ell})$ on real training samples. A test sample passes if:

$$(\ell - \mu_{\ell})^T \Sigma_{\ell}^{-1} (\ell - \mu_{\ell}) \leq \chi_{0.95}^2(3) \quad (14)$$

where $\chi_{0.95}^2(3)$ is the 95th percentile of the chi-squared distribution with 3 degrees of freedom.

Stage 2 Filter (Latent Space): We extract latent vectors by temporal averaging:

$$\bar{\mathbf{z}}_b = \frac{1}{T} \sum_{t=1}^T \mu_b(\cdot, t), \quad \mathbf{z} = [\bar{\mathbf{z}}_{\text{LF}}, \bar{\mathbf{z}}_{\text{BP}}, \bar{\mathbf{z}}_{\text{HF}}] \in \mathbb{R}^{36} \quad (15)$$

We apply PCA to reduce dimensionality to 10D (preserving $> 91\%$ variance) and fit $\mathcal{N}(\mu_z, \Sigma_z)$ on real samples. A

sample is classified as real only if it passes both filters:

$$\text{Prediction} = \begin{cases} \text{Real} & \text{if Stage 1 pass AND Stage 2 pass} \\ \text{Fake} & \text{otherwise} \end{cases} \quad (16)$$

This approach provides robustness: reconstruction filters gross anomalies, while latent filtering captures subtle distribution shifts.

3. Experiments

3.1. Dataset and Setup

In our implementation, Stage 1 is trained on a large-scale set of **35,186 live video samples**, providing a strong one-class prior over genuine facial dynamics.

Fake videos deviate from this learned distribution, enabling anomaly detection.

In stage 2, We collect a dataset of facial videos containing:

- **Training:** 1088 real, 538 fake
- **Testing:** 117 real, 117 fake

Videos are recorded at 30 fps with 10-second duration. We use 40 facial landmarks (20 outer lip, 20 inner lip contour) extracted via a pre-trained detector.

All models are trained with Adam optimizer ($\text{lr} = 10^{-4}$), batch size 64, for 20 epochs (Stage 1) and 10 epochs (Stage 2). We use mixed precision training on a single GPU.

3.2. Proof-of-Concept: One-Class TCN-VAE Pipeline

Before developing the full frequency-decoupled architecture, we implemented a proof-of-concept (PoC) system based on a single-band one-class TCN-VAE trained on lip landmark sequences.

Table 1. Performance comparison on test set (234 samples). All methods use two-stage Gaussian filtering for inference.

Method	Acc	Prec	Rec	F1	Spec
Poc	70.05	62.61	99.55	76.87	40.54
Stage 1 Pretrain	85.90	82.31	91.45	86.64	80.34
Method 1: Margin	88.46	85.71	92.31	88.89	84.62
Method 2: Discrim.	88.89	86.40	92.31	89.26	85.47

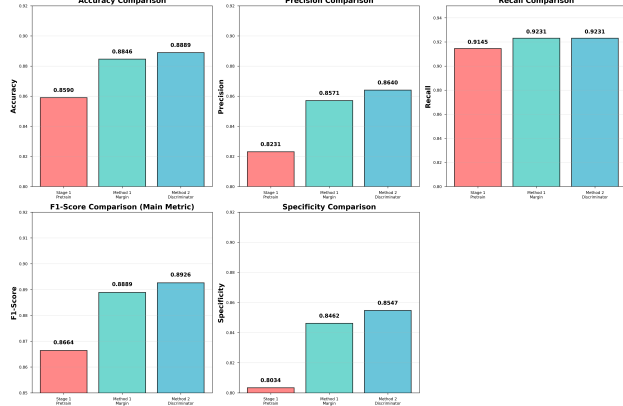


Figure 2. Performance comparison across all metrics. Method 2 (Discriminator) achieves the best F1-score of 89.26%, slightly outperforming Method 1 (88.89%) and significantly exceeding Stage 1 baseline (86.64%).

One-class TCN-VAE architecture. The model operates on preprocessed lip landmark time series, using the same temporal convolutional backbone as the band-split VAE. Concretely, we use a TCN-based encoder-decoder pair:

- **Encoder (TCNEncoder):** a stack of dilated 1D convolutional blocks that maps the input sequence to latent parameters ($\mu, \log \sigma^2$).
- **Decoder (TCNDecoder):** a symmetric TCN that reconstructs the original sequence from latent samples.

3.3. Main Results

Table 1 compares our three models on the 234-sample test set.

Figure 2 visualizes the performance across all metrics.

Key Observations:

(1) **Both Stage 2 methods achieve strong performance**, with Method 2 (Discriminator) slightly outperforming Method 1 (Margin) by 0.37% F1. This suggests that with limited fake data, both approaches are effective, with the discriminative approach offering marginal gains. Both significantly outperform Stage 1 baseline (+2.25% and +2.62%

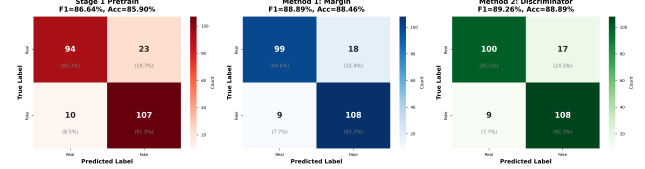


Figure 3. Confusion matrices for all methods. Method 2 achieves TN=100, FP=17, FN=9, TP=108. Both Stage 2 methods significantly reduce false negatives compared to Stage 1.

F1 respectively), demonstrating the value of discriminative fine-tuning.

(2) **High recall (92.31%)** indicates our models rarely misclassify fake samples as real—critical for security applications where false acceptance is costly.

(3) **Balanced specificity (84.62–85.47%)** shows the models maintain good real sample acceptance while detecting fakes, avoiding overly conservative rejection.

(4) **Small fake dataset suffices:** Using only 538 fake samples (vs. 1088 real), we achieve near-90% F1, validating our hypothesis that extensive fake datasets are unnecessary with proper one-class pretraining.

Figure 3 shows confusion matrices for all three methods.

3.4. Training Dynamics

Stage 1 focuses on learning a compact representation of real facial motion, with reconstruction loss converging after roughly 15 epochs. Stage 2 builds upon this foundation, quickly learning to distinguish fake samples with minimal additional training and stabilizing near the best F1-score of 89.26% by epoch 10.

3.5. Frequency Band Analysis

The frequency decomposition provides interpretability. Analysis of reconstruction errors reveals that fake videos exhibit characteristic patterns:

- **LF anomalies:** Unnatural head pose transitions in replay attacks due to camera shake and environmental motion
- **BP anomalies:** Asynchronous lip motion in deepfakes where mouth movements don’t perfectly match speech frequencies
- **HF anomalies:** Missing or suppressed micro-expressions in synthetic faces which lack subtle muscle movements

This interpretability aligns with prior work that exploits

physiological or biological signals (e.g., remote photoplethysmography) for deepfake detection (Ciftci et al., 2020), but our approach focuses on frequency-structured landmark motion rather than color-based PPG signals.

3.6. Computational Efficiency

Our model requires only 2.7M parameters and processes 10-second videos.

In comparison:

- Video-based 3D CNNs: 100M parameters,
- Transformer-based methods: 80M parameters,

The parameter efficiency (38 times smaller than video-based approaches) enables deployment on resource-constrained devices such as mobile phones and embedded systems.

4. Discussion

4.1. Key Contributions

Our work demonstrates that effective liveness detection is achievable with minimal supervision and computational resources. The three main contributions—landmark-only representation, two-stage learning, and frequency decomposition—synergistically address the limitations of existing methods. Beyond the face anti-spoofing setting, our design choices are closely related to trends in deepfake video detection. Recent surveys have highlighted the need for detectors that are both computationally efficient and robust to evolving manipulation techniques (Yu et al., 2021; Kaur et al., 2024). Landmark-only, frequency-decoupled representations provide a complementary direction to biological-signal-based approaches such as FakeCatcher (Ciftci et al., 2020) and audio-visual mismatch detectors for lip-sync deepfakes (Bohacek & Farid, 2024), suggesting that combining multiple lightweight cues (physiological, geometric, and semantic) may further improve robustness.

Landmark efficiency: Operating on 40 landmarks instead of full frames reduces data dimensionality by 1000 times while preserving essential motion dynamics. This choice is motivated by the observation that facial motion patterns, rather than appearance details, are the primary distinguishing features between real and fake videos. Our results validate this hypothesis: 89.26% F1-score with 2.7M parameters demonstrates that motion alone suffices for high-accuracy liveness detection.

Data-efficient learning: The two-stage paradigm addresses a fundamental challenge in liveness detection: obtaining diverse fake samples is expensive and quickly outdated as attack methods evolve. By learning the real mani-

fold first through one-class VAE, then fine-tuning with minimal fake data (538 samples), we show that extensive fake datasets are unnecessary. Stage 1 provides strong initialization by capturing the structure of genuine motion; Stage 2 adds discriminative boundaries with minimal additional data.

4.2. Limitations and Future Work

Landmark dependency: Our method assumes reliable landmark detection. While modern detectors are robust, extreme poses or occlusions may degrade performance. Future work could incorporate uncertainty estimation from the landmark detector or develop fallback mechanisms for low-confidence regions.

Cross-dataset generalization: All experiments use a single dataset captured under controlled conditions. Cross-dataset evaluation would assess generalization to different capture devices, lighting conditions, demographics, and compression artifacts.

5. Conclusion

We presented an efficient two-stage framework for face liveness detection that operates exclusively on facial landmarks. By combining one-class VAE pretraining with minimal discriminative fine-tuning and frequency-decoupled architecture, we achieve 89.26% F1-score with only 2.7M parameters—38 times smaller than video-based approaches.

Our results challenge the conventional wisdom that liveness detection requires extensive labeled fake datasets and computationally expensive models. With proper inductive biases (landmark motion, frequency decomposition) and learning strategies (one-class pretraining), competitive performance is achievable with minimal resources. The frequency decomposition not only improves accuracy but also provides interpretability, enabling forensic analysis and targeted defense against evolving attacks.

Future work will explore cross-dataset generalization, robustness to emerging attack methods (diffusion-based deepfakes, adversarial perturbations), and deployment optimizations for real-time systems. We believe this work opens new directions for efficient and interpretable biometric security.

References

- Bohacek, M. and Farid, H. Lost in translation: Lip-sync deepfake detection from audio-video mismatch. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 4315–4323, 2024. doi: 10.1109/CVPRW63382.2024.00435.

- Ciftci, U. A., Demir, I., and Yin, L. Fakecatcher: Detection of synthetic portrait videos using biological signals. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2020. doi: 10.1109/TPAMI.2020.3009287.
- Kaur, A., Noori Hoshyar, A., Saikrishna, V., Firmin, S., and Xia, F. Deepfake video detection: Challenges and opportunities. *Artificial Intelligence Review*, 57(6):1–47, 2024. doi: 10.1007/s10462-024-10810-6.
- Yu, P., Xia, Z., Fei, J., and Lu, Y. A survey on deepfake video detection. *IET Biometrics*, 10(6):607–624, 2021. doi: 10.1049/bme2.12031.